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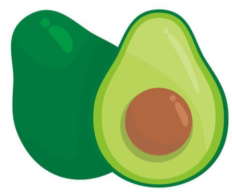

“Green Gold”: Avocado Production and its implications on Global Water Scarcity

B141395

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Avocado orchard in Mexico (Gary Coronado / Los Angeles Times)

1. Introduction

1.1 Rise in Avocado consumption

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Avocado (Persea americana) is a warm climate tree species native to Mexico, whose fruit is characterized by its unique texture, flavor and nutrients, and can be used for direct consumption, as well as for production of various products including oil and cosmetics (Schaffer et al., 2013). Avocado consumption is increasingly popular amongst countries in Europe, North America and the developed parts of the world in recent decades. It is estimated that 5 million tonnes of avocados are consumed each year, and on average 3.5 and 1.05 kg per capita in the USA and Europe respectively (Sommaruga et.al, 2020).

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In Europe, avocado consumption per capita had increased by 179% on average from 2012 to 2018 and it is still expected to rise.

-(Sommaruga et.al, 2020)

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1.2 Waste of Water in Avocado Production

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A significant portion of this demand increase is driven by younger generations in developed countries drawn by the potential health benefits of avocados, and the rise of flexitarian diets trends. Plant-based foods are generally better for the environment than animal products, however, not all plant-based food products have a small environmental footprint.

Fruits and vegetables have the highest average water footprint per calories when compared with other types of crops (excluding spices and nuts) (Mekonnen and Hoekstra, 2011). Adapted to hot and humid rainforest habitats of Southern Mexico, avocado trees have shallow roots that are poor at searching out water held within the soil. In subtropical, tropical and Mediterranean climates where commercial production of avocado usually takes place, avocado trees require substantially more water than other crops and cannot be grown at commercial scale without supplementary irrigation (Sommaruga et.al, 2020). A staggering 140 litres to 272 litres of water will be needed to grow a single avocado, or about 834 litres per kilogram of avocado (Frankowska et.al, 2019; Frank, 1983). The huge water consumption puts a huge pressure on the already water-stressed regions such as California, Chile, Mexico and southern Spain, where most commercial avocado crops are grown.

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> Growing a kilogram of avocados requires 834 litres of water, while the same amount of plums only needs 305 litres
>
> -(Frankowska et.al, 2019)
>
> ”

Mexican police confiscating Avocados from cartels that have evolved beyond drug trafficking and now prey on the avocado trade.(LatinAmerican Post)

1.3 Inequality in water distribution caused by Avocados

In areas such as Peru and Chile, the increasing demand of avocados has led to illegal extraction from rivers as violent cartels and large cooperatives get involved in avocado agribusinesses and has been accused of creating water conflicts and stresses in those areas. In Chile, water rights are privatised, and water is regulated by private property law under the Chilean water code. Large avocado orchards have legally been able to empty the rivers, and as a result, the rivers have dried up, and local residents' human right to water is being violated.

Global economic forum (2014) had established water resource availability as one of the major global risks threatening our future sustainability (Musolino et.al, 2018). As agriculture irrigation accounts for around 70% of the water use worldwide as of now (OECD, 2017), managing the potential imbalances in agricultural water uses is essential to maintain sustainable water availability.

2. Current methods

2.1 Importations

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In the UK, 100% of the avocados are imported (DEFRA, 2015; ITC, 2018). The subtropical Mediterranean climate of southeastern Andalusian coast allows Spain to be the largest European avocado producer (Moreno-Ortega et al., 2019). The avocado agribusinesses in Spain are generally better regulated and managed than their South American counterparts, although water consumption is still an issue, there are less inequalities in distribution of water.

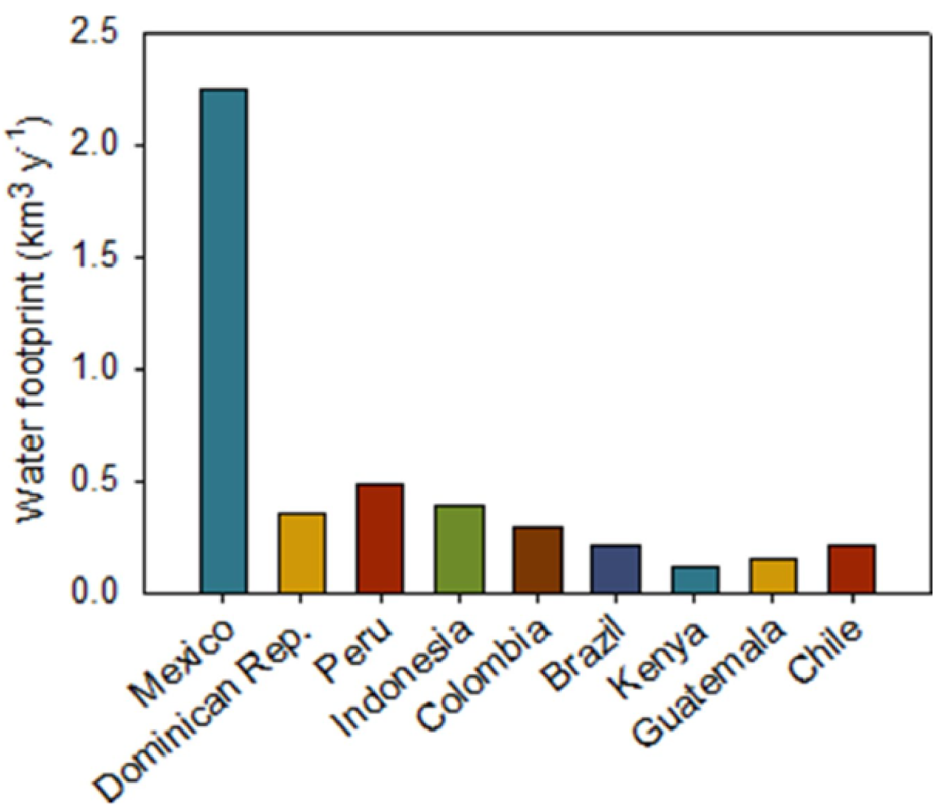


Figure 1: *Total water footprint corresponding to the annual production of avocado in 2018 for the nine top avocado producers (Sommaruga et.al, 2020)*

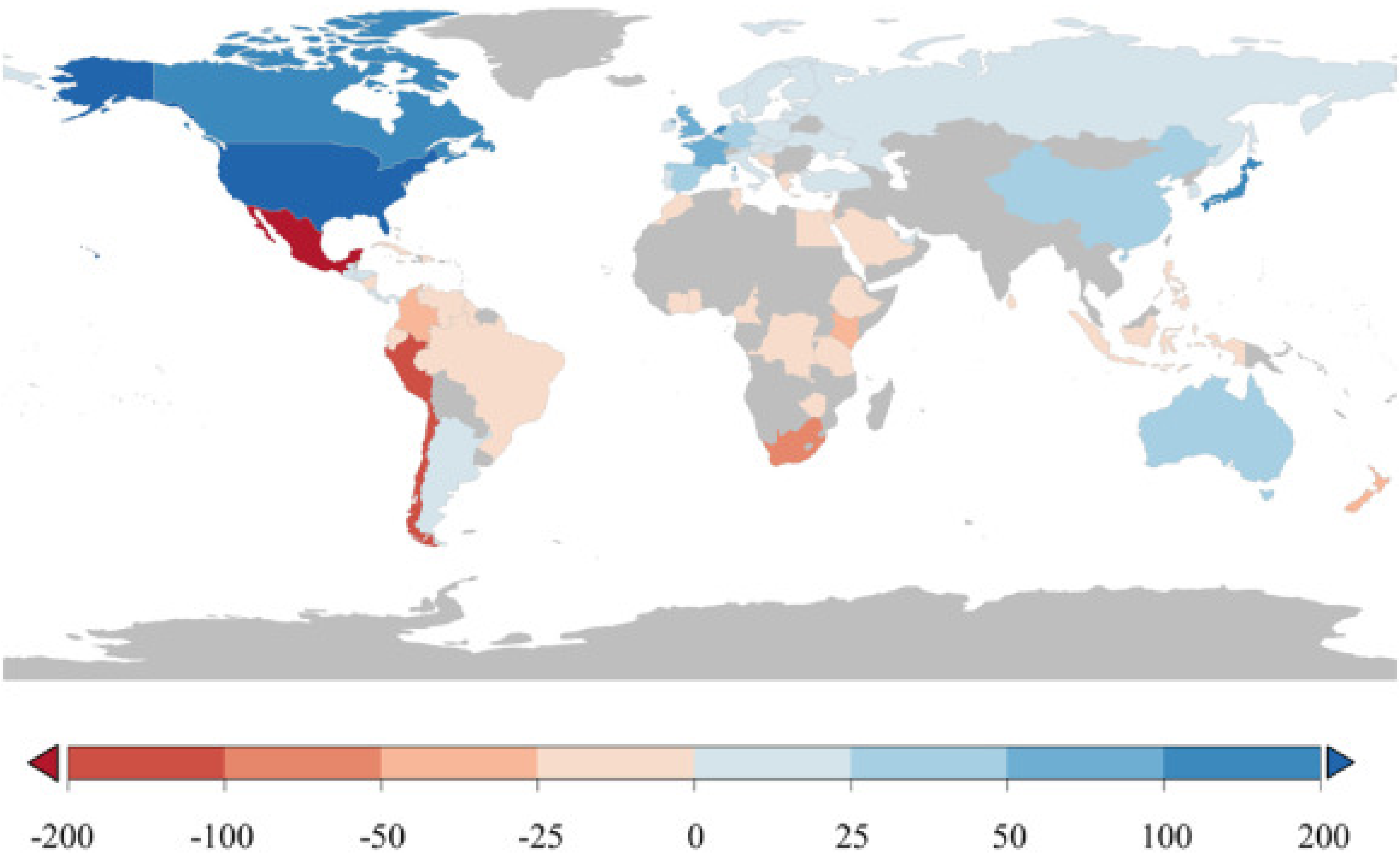


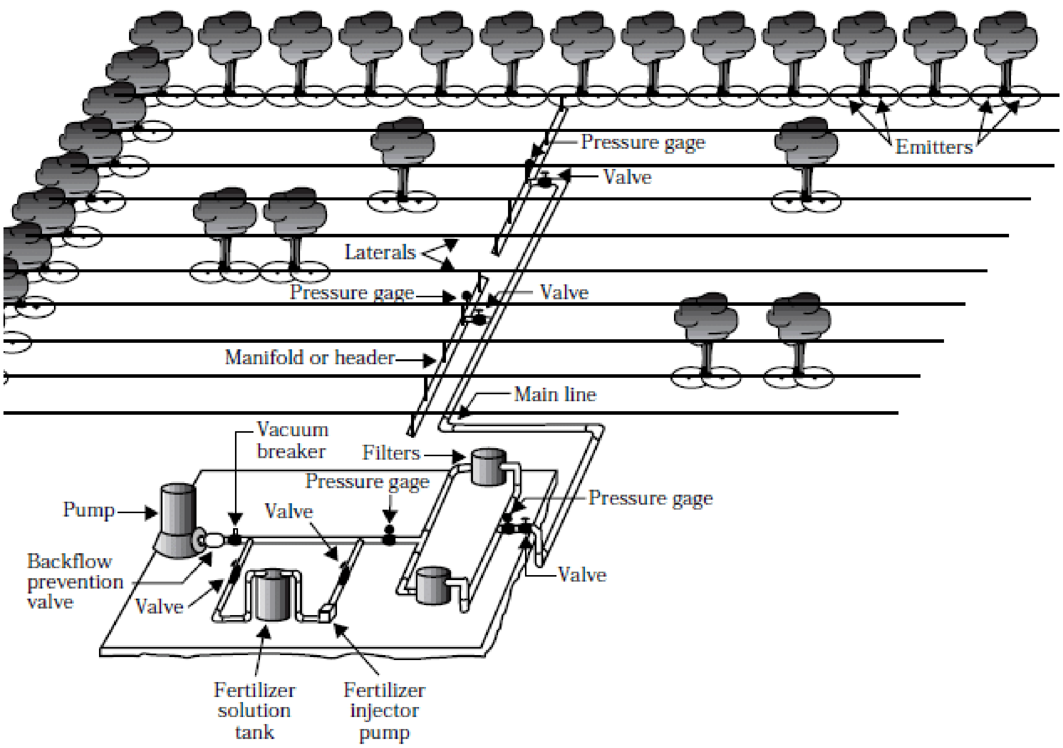
***Figure 2** The net virtual water import (Mm^3) of countries in 2016. Net virtual water exporters result in negative values and net virtual water importers result in positive values.*

However, in April 2020, the EU signed an updated free trade agreement with Mexico which allows tariff-free import of Mexican avocados into the EU, increasing the share of Mexican avocados in the EU market. Mexican avocados had the highest water footprint in 2018 (FAOSTAT 2021 ; Figure 1) and their water consumption had doubled in just two decades (Oswald Spring, 2011). The increased demand from Europe could lead to further exporting of virtual water from South America (Figure 2), worsening the water conflicts between local communities.

2.2 Using Drip Irrigation Systems

Drip irrigation systems are used to avoid waste of water and ensure homogeneous distribution of water so that each tree receives exactly the amount it needs. Tensiometers are used to measure the humidity of the soil which indicates the exact time irrigation should start and stop. Most governments and farmers are increasingly aware of the water waste in agricultural uses and are endeavouring to promote efficient agriculture through improved irrigation management.

Adoption of plastic mulch covering the raised beds of young avocado trees can directly protects the area irrigated by the dripper, slowing the evaporation of water which would retain soil moisture and provide better water availability for the plant.



Typical drip irrigation system layout (Usmonov and Gregoretti, 2020)

Terrace Avocado Orchards (Mission Kwasizabantu, 2020)

2.3 Controlled Extraction from Aquifers

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In Spain, extraction of water from the subsoil aquifers is regulated by the government and water meters are used in farm wells to limit the amount of water extracted according to the reserve of the subsoil. Farmers are made aware the recharge capacity of aquifers should not be over-exploited since most avocado orchards in Spain located near the coast, penetration of sea water can turn aquiders saline and make them permanently unsuitable for irrigation as salt will burn the plant's roots. Farmers therefore tend to set up terrace avocado orchards where they cultivate terraces on slopes to increases the infiltration rate of water, thus replenishing the aquifers and reducing runoffs.

3. Policy Recommendations

3.1 Road Map

It is undoubted that increased demand of avocados has led to water stresses in producing countries. A road map is needed to bring together governments, large and small producers, as well as local communities near avocado orchards that suffer from water conflicts to develop a strategic action plan addressing water scarcity and availability problems caused by the production of avocados and other water intensive crops and look for sustainable production methods with lower water usage.

Population growth, and climate change put huge pressure on the global water availability, with a particular impact on agriculture. Population growth is expected to increase to over 9 billion by 2050 (FAO, 2012) and this extra population will need food to meet its basic needs, as a result, food demand is expected to increase by 60% (Sauer et al., 2008). It is important for this road map to assess the critical capacity of production of such crops and consider future climate change scenarios to maintain sustainable agricultural water usage to support nutritionally adequate human diets in the future.

Large agribusinesses with higher capacity to respond to water stress conditions should not endanger small farmers, and should hold greater responsibility for the downstream impact of their water extractions. Major importing countries should also consider the environmental footprints of avocados when deciding where to import and promote imports from countries where current and future water scarcity is not further endangered.

Human capital variables, such as, owner age and education make a big difference in grower decisions in adoption of water-saving technologies and management practices (Engler et. al 2016). It is therefore important to educate farmers to make the conscious effort in conserving water and be aware of the consequences of water waste besides the economic costs. Consumers should also be educated and encouraged to engage in ethical consumption of food products, to protect the sustainability of communities in producing areas.

3.2 Plastic Covers as makeshift greenhouses

Plastic covers can be raised over the canopy of the trees in less developed avocado producing countries with less access to technologies. Creating essentially a low cost greenhouse that reduces water demand and increases yields. It can change the micrometeorological conditions that affect the growth of crops. Inside these made-shift greenhouse, the temperature increases by between 1 and 2 °C, the relative humidity also increases and the vapour pressure deficit falls. It also reduces the amount of radiation and wind.

3.3 Sustained Deficit Irrigation

Sustained deficit irrigation (SDI) can provide a new approach to water savings in avocado production. It involves reducing irrigation supply below the maximum evapotranspiration requirements which decreases productivity (English, 1990; Elias et l., 2017).

Although SDI sounds counter-intuitive as it reduces productivity and crop size, it optimizes the economic output in water scarcity scenarios, emphasizing Water productivity and prioritises net income per unit water used rather than per land unit. Further studies on management of deficit irrigation specifically on avocados are necessary for sustainable and profitable avocado production.

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